# THIJMEN WEIJGERTZE

GRADUATED IN LIFE SCIENCES

## PERSONALIA


**Adress**
<Redacted>

**Date of Birth**
- 27-01-2003

**Contact**
- <Redacted>
- thijmen.lifesciences@gmail.com

**Links**
- Data Science portfolio
- GitHub
- LinkedIn


**Driving License**
Type B

## WORK EXPERIENCE

**Utrecht University (Internship)**
- Period: September 2024 – July 2025 (10 months, 38 hours/week)
- Techniques: BMDC cell-culture, qPCR, conventional flow cytometry, cell culture and T-cell recognition assays

## TOP 5 SKILLS

- Helpful
- Reliable
- Passionate
- Eager to learn
- Detail-oriented

## HOBBIES

- Piano improvisation
- Programming (web hosting, Linux VM management and R)
- Social activities with friends
- Travel photography
- Microscopy
- Cooking

## ABOUT ME

Graduated in Life Sciences at the University of Applied Sciences Utrecht, specializing in Biomolecular Research, with a 10-month graduation internship at Utrecht University. Experienced in various laboratory techniques, literature research and reporting of scientific results. Additionally, foundational knowledge of data analysis through a minor in Data Science. Interested in research positions that combine experimental work with literature study and data interpretation.

## DEGREES

**HAVO-degree, Marnix College Ede**
Sep 2015 – May 2020
- Curriculum: "Nature & Technology" and "Nature & Health"

**Bachelor Biology and Medical laboratory research (Life Sciences), HU University of Applied Sciences Utrecht**
September 2020 – July 2025

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|---|---|
| - Specialised in: | Biomolecular Research (BMR) |
| - Minor in: | Data Science for Biology - HU |
| - Internship at: | University of Utrecht (10 months) |
| - Diploma level: | NLQF 6 / EQF 6 |

## PRACTICAL SKILLS

**Lab**
Agarose gel electrophoresis, ÄKTAprime, Calcium phosphate transfection, C. elegans, cDNA synthesis, Cell culture, Conventional flow cytometry, Differential centrifugation, ELISA, Light, dark & fluorescence microscopy, Molecular cloning, Paper electrophoresis, (q)PCR, RNA/DNA isolation, Safe microbiological techniques, SDS-PAGE, SEC & TLC chromatography, Sonication, Spectrophotometry, Western blot

**Data Science**
R, Rmarkdown, Rbookdown, Rpackage, Git/github, Bash, SQL*, Python*, Machine Learning*, frequentist statistics, Reproducible research, Data Science portfolio
* Basic knowledge

## THEORETICAL KNOWLEDGE

**Theoretical knowledge within the subjects:**

- Molecular biology
- Tumor cell biology
- Immunology
- Genetics
- Advanced lab tools (incl: FACS, NGS, proteomics, MS)
- Experimental Design & Cell Culture
- Data science
- Cellular Biochemistry
- Biotechnology
- Pharmacology
- Haematology
- Physiological Regulation systems
- Metabolism and muscle physiology

## MAIN PHASE PROJECTS

**Lipid nanoparticle-induced endosomal rupture (internship)**

- Utrecht University – 10 months, 38 hours/week – English
- Investigated endosomal disruption by LNPs (ALC-0315, SM-102) in dendritic cells (BMDCs & DC2.4) in the context of autoimmune treatments
- Analyzed: galectin and IL-6 gene expression, functional mRNA delivery and surface expression (CD40, CD86, MHCII)
- Demonstrated significant differences between DC models and LNP formulations
- Techniques: BMDC cell culture, conventional flow cytometry, qPCR, T-cell recognition assays

**Practicum Data Science (minor)**

- University of Applied Sciences Utrecht – 6 months – Dutch/English
- Developed: a bioinformatics pipeline commissioned by RIVM for detecting resistance genes in plasmids
- Pipeline in bash: fastq-dump, FastQC, Trimgalore, Unicycler, PlasmidEC, gplas2 and ABRicate
- Applied: GitHub workflow, Bash, version control of packages and vignette documentation

**Practicum Biomolecular Research (specialization)**

- Worked on the production and isolation of HPO lyase from an E. coli expression system
- Scaled up bacterial suspension to 80 mL and induced with IPTG
- Protein isolated via sonication, centrifugation and ÄKTAprime; measured with spectrophotometry
- Novel gene isolated from garden cress via RNA isolation, cDNA synthesis and two-step PCR

**Practicum Pharmacology**

- Investigated the effect of N-acetylcysteine (reduced mucus layer) on susceptibility to S. Typhimurium infection in C. elegans
- Infection measured with qPCR and CFU determination
- PAS staining protocol developed and documented for follow-up research

**Practicum Microbiology**

- Investigated a potential influenza A (H1N1) lung infection and bacterial co-infection in a patient case
- Influenza detected via cDNA synthesis and real-time PCR
- Bacterial infection examined through identification and disk diffusion (antibiotic resistance)
- End product: extensive literature review and diagnosis (group G streptococcus / S. epidermidis)

**Practicum Genes and Proteins**

- Designed: a functional pAcGFP-IL-8 DNA construct for research into IL-8 gene expression in RSV infection
- IL-8 inserts isolated from plasmids and inserted into pAcGFP1 vectors
- Techniques: molecular cloning, SDS-PAGE, western blot, agarose gel electrophoresis, fluorescence microscopy